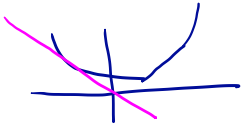
Summary of Gradient and Subgradient Method

min: $f(x)$

f convex: local opt $\Rightarrow$ global optimality



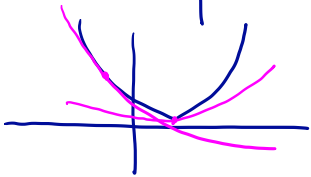
f can be globally approx'd from below by linear functions.

f smooth + convex:

f has a quadratic upper bound.

$\Rightarrow$ self-tuning property $\nabla f(x) \to 0$ as $x \to x^*$.

f strongly convex



f has a quadratic lower bound.

$\Rightarrow$ $\nabla f(x)$ is large when x is far from x^*.

Summary of Gradient and Subgradient Method

f convex: subgradient method: $O(1/\varepsilon^2)$

$$\updownarrow$$

$$O(1/\sqrt{T})$$

f convex & smooth: gradient descent: $O(1/\varepsilon)$

f strongly convex: subgradient method: $O(1/\varepsilon)$

f smooth & strongly convex: gradient descent: $O(\log(1/\varepsilon))$